Lab 4 — Submission Setup, and Object Oriented Programming

git, from scratch, and your first push — then the OOP exercises

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Today

Two halves

- **Part 1 — Submission setup.** Fork, clone, wire the remotes, push. We do this together, from scratch, right now. **Lab 2 is due tonight, and you submit it here.**
- **Part 2 — Object Oriented Programming.** The lab itself: 12 exercises, using the Lectures 8–9 material.

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- `Labs/GitHub-Setup.pdf` — the same steps as Part 1, written out line by line, with a table of what to do when git complains
- `Labs/Lab.4/Lab.4.ipynb` — everything below, plus the exercises, in one notebook

git — what it is, and the words you'll use

What git is

- A **version control system** — it tracks how files change over time
- Built for code, good for any large document
- Lets several people work on one thing without overwriting each other
- Lets you keep parallel versions, and mark releases
- Works entirely on your own machine, or against a server

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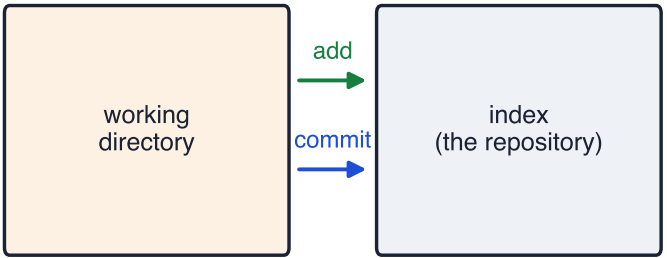
- **GitHub** runs a public git server, and has become how open-source code is shared
- **GitLab** is a similar thing companies often host themselves

A repository holds two things



a clone holds
both halves

- `add` puts a file's changes in line to be recorded
- `commit` records them, with a message saying what you did

The words you'll see

- **Clone** — your own local copy of a repository. What you actually work in
- **Remote** — a copy living on a server, that your clone talks to
- **Fork** *(a GitHub feature, not a git command)* — a copy of someone else's repository on *your* GitHub account, that can now go its own way
- **Branch** — a parallel version that doesn't disturb the main one (`main`)
- **Tag** — a name pinned to one particular version

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- **fetch** — get changes from a remote · **merge** — fold them into your files
- **pull** = fetch + merge · **push** — send your commits to a remote
- **Merge conflict** — when two people changed the same thing and git can't decide. Whoever pushes second resolves it *(there's a trick to avoid this in this course entirely — next section)*
- **Pull request** — asking to pull *your* fork's changes back into someone else's repo. Not something you'll do in this course — we only push to our own fork — but you'll see the term everywhere `git` is used for real

Part 1 — Submission Setup

Why a fork, specifically

- The course repository is **public** — `UTA-DataScience/DATA3402.Fall.2026`
- You don't have write access to it, so you need your own copy to push to — that's what the fork is for

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- The course repository is **public** — `UTA-DataScience/DATA3402.Fall.2026`
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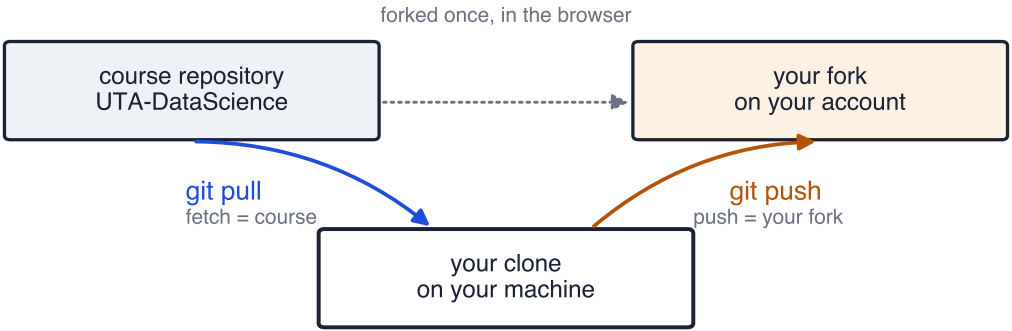
- Because the class repo is public, **your fork will be public too**. That's expected.

How this course is wired



one remote, named origin — it fetches from one place and pushes to another

Pull from class. Push to my fork. Never force push.

Setup — once per semester

1. Fork the class repo, in your browser
2. `git clone` your fork onto your machine
3. `cd` into the clone
4. `git remote remove origin`
5. `git remote add origin <class repo URL>`
6. `git remote set-url --push origin <your fork's URL>`

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Check it: `git remote -v` should show the **class repo for fetch**, and **your fork for push**. If both lines show the same URL, step 6 didn't take — run it again.

Every week

every lab, the same five steps

Avoid the merge conflict before it happens

The moment you `git pull` a new lab, copy it to a new name and work there — never edit the file we handed you.

```
cp Lab.2.ipynb Lab.2.solution.ipynb
```

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```
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```

- `git pull` only touches files it recognizes from the class repo. `Lab.2.ipynb` is one of those; `Lab.2.solution.ipynb` isn't — we never handed out a file by that name
- So a later `git pull`, even one that updates `Lab.2.ipynb` after we fix something in it, **can never conflict with your copy.** There's nothing shared left to disagree about
- Do this **first**, before you write a line of code — not at submission time. If you've already been editing `Lab.2.ipynb` directly, copy it now and switch to the copy

Submitting

1. `git pull` — get this week's material
2. Confirm you've been working in `Lab.2.solution.ipynb`, not `Lab.2.ipynb` — if not, copy it over now, before you do anything else
3. `git add Labs/Lab.2/Lab.2.solution.ipynb`
4. `git commit -a -m "Lab 2 submission"`
5. `git push`
6. Check GitHub — your fork should show the new commit

If it goes wrong

- `rejected (non-fast-forward)` on your very first push — expected. Your fork and the class repo have separate histories: `git config pull.rebase false`, pull, then push again
- **Merge conflict** — you and the class both changed the same file, and git can't decide which version wins. `Labs/DATA3402_Lab3_Merge_Conflict_Guide.pdf` walks through resolving one, including inside a notebook's JSON
- Full error table, line by line: `Labs/GitHub-Setup.pdf`

Before you leave today

- Confirm your fork shows the pushed commit on GitHub
- Need to keep working later? Edit the same clone, then repeat add/commit/push

Part 2 — Object Oriented Programming

From Lectures 8–9

- **Encapsulate** — bundle data and the operations on it into one object
- **Public / private** — by convention, a leading `__` means "internal to the class"; everything else goes through accessor functions you write
- **Inheritance** — a base class states the common interface; subclasses fill it in
- `Canvas, Shape, RasterDrawing` — the drawing classes from Lecture 9 are exactly what exercises 10–12 build on

The arc of the 12 exercises

- **1–2** — a counter class, then the same thing with private data
- **3–6** — rectangle, circle, a shared base class, triangle
- **7–9** — perimeter points, point-containment, overlap
- **10–12** — paint it: a `Canvas`/`paint` module, `RasterDrawing`, save/load

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Each exercise builds on the last. A shaky base class in 5 doesn't just break 5 — it makes 6 through 9 harder too. Get help early rather than push on with it.

Save and load, the trick

- Overload `__repr__` to return the **Python code that would reconstruct the object**
- Saving is writing that string to a file
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`Labs/Lab.4/Lab.4.ipynb` has a worked `foo` / `foo_loader` example at the very end — read it before you start exercise 12, not after.